power

File: power.kicad_sch

5 V analog / 3.3 V rails, 2.5 V reference, V_cm buffer, current-mirror reference

line0

File: line0.kicad_sch

line 0: RC + buffer, 56-section LC ladder, 8 taps

line1

File: line1.kicad_sch

line 1: RC + buffer, 56-section LC ladder, 8 taps

line2

File: line2.kicad_sch

line 2: RC + buffer, 56-section LC ladder, 8 taps

line3

File: line3.kicad_sch

line 3: RC + buffer, 56-section LC ladder, 8 taps

line4

File: line4.kicad_sch

line 4: RC + buffer, 56-section LC ladder, 8 taps

line5

File: line5.kicad_sch

line 5: RC + buffer, 56-section LC ladder, 8 taps

line6

File: line6.kicad_sch

line 6: RC + buffer, 56-section LC ladder, 8 taps

line7

File: line7.kicad_sch

line 7: RC + buffer, 56-section LC ladder, 8 taps

crossbar

File: crossbar.kicad_sch

8x8x8x2 footprints: W=+1 -> RXP, W=-1 -> RXN, W=0 -> both DNP

ch0

File: ch0.kicad_sch

output 0: summers, bias, BJT tanh, diff-amp, S/H

ch1

File: ch1.kicad_sch

output 1: summers, bias, BJT tanh, diff-amp, S/H

ch2

File: ch2.kicad_sch

output 2: summers, bias, BJT tanh, diff-amp, S/H

ch3

File: ch3.kicad_sch

output 3: summers, bias, BJT tanh, diff-amp, S/H

ch4

File: ch4.kicad_sch

output 4: summers, bias, BJT tanh, diff-amp, S/H

ch5

File: ch5.kicad_sch

output 5: summers, bias, BJT tanh, diff-amp, S/H

ch6

File: ch6.kicad_sch

output 6: summers, bias, BJT tanh, diff-amp, S/H

ch7

File: ch7.kicad_sch

output 7: summers, bias, BJT tanh, diff-amp, S/H

connectors

File: connectors.kicad_sch

J_IN, J_OUT 2x13, J_PS 2x8, 2 x MCP4728, tile ID, monitor

model tdlnet_b_w8_aware2_fro_ternary.npz R_f=976 R_b=7870 (derived 20000) designed, not built

dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /

File: tile.kicad_sch

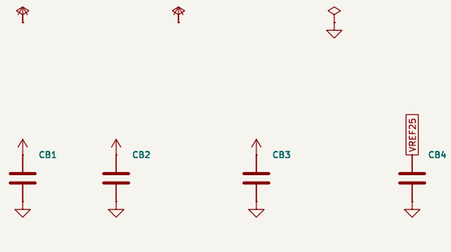
Title: TILE-8 layer-3 variant (M3a)

Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 1/20

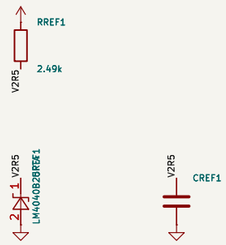
TILE-8 power: +5VA (analog, from ribbon), +3V3 (logic reference, pass-through), 2.5 V V_cm reference, current-mirror reference for the 8 diff-pair tails

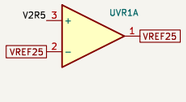
Rails enter on the ribbon (J_IN pins 17,19 = +5VA, 21 = +3V3, GND) and pass to J_OUT

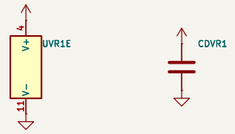
LM4040B25 2.5 V shunt reference (1 mA via 2.49k) -> TLV9064 follower -> VREF25. V_cm buffer = unit 1 of UVR



CB1,CB2: 10u +5VA CB3: 10u +3V3 CB4: 10u V_cm (VREF25)







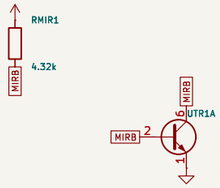
10k pull-down on SH_CLK



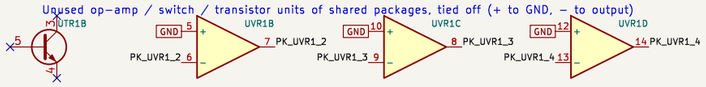
Input-buffer quad power (the 8 line input buffers share two quads; units placed on the line sheets)



Mirror: UTR unit1 diode-connected + R_ref 4.32k -> 1.008 mA reference; tails are the 8 units UT0..UT3 on ch sheets



Unused op-amp / switch / transistor units of shared packages, tied off (+ to GND, - to output)



dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /power/
File: power.kicad_sch

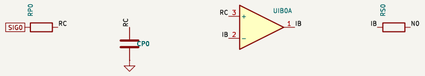
Title: Power, reference, tail-mirror reference

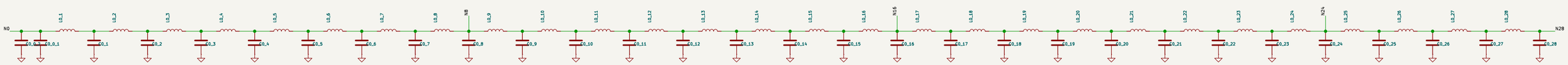
Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 2/20

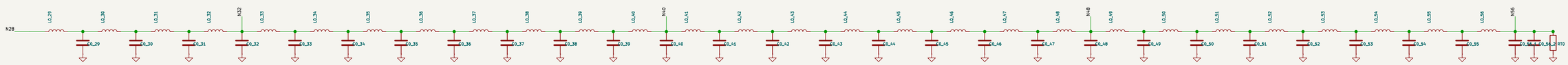
Delay line 0: 56 x (L 120 uH FHW1812IF121JST + C 120 pF C0G), Z0 = 1 kohm, T = 120 ns/section, taps k = 0..7 at sections 0, 8, ..., 56 (T_s = 8 sections = 0.96 us)

All LO_s: 120 uH 5 %, 1812; CO_n: 120 pF C0G 0603; end caps 2 x 30 pF (60 pF) at nodes 0 and 56; Rs = Rt = 1k

RP+CP: 1k + 330 pF = 482 kHz input RC; RS = source termination 1k

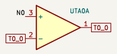




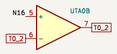


Tap buffers: TLV9064 followers, outputs TO_k feed the crossbar (global labels)

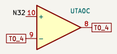
tap 0 (node 0)



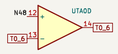
tap 2 (node 16)

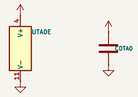


tap 4 (node 32)

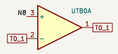


tap 6 (node 48)

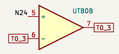




tap 1 (node 8)



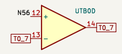
tap 3 (node 24)

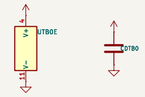


tap 5 (node 40)



tap 7 (node 56)





dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /line0/
File: line0.kicad_sch

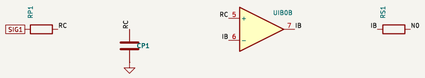
Title: Delay line 0

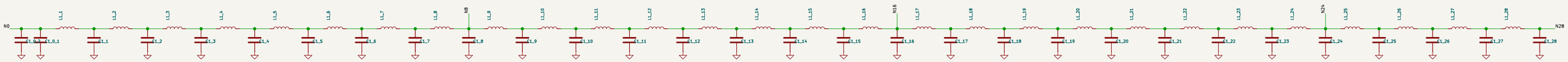
Size: A2	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 3/20

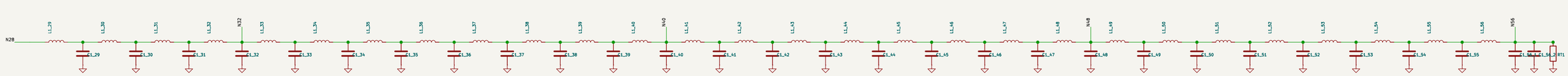
Delay line 1: 56 x (L 120 uH FHW1812IF121JST + C 120 pF C0G), Z0 = 1 kohm, T = 120 ns/section, taps k = 0..7 at sections 0, 8, ..., 56 (T_s = 8 sections = 0.96 us)

All L1_s: 120 uH 5 %, 1812; C1_n: 120 pF C0G 0603; end caps 2 x 30 pF (60 pF) at nodes 0 and 56; Rs = Rt = 1k

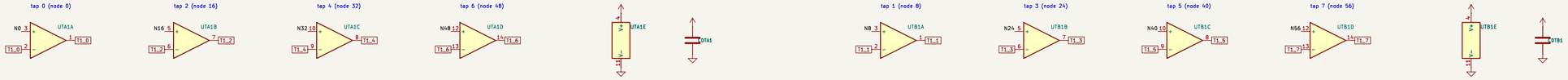
RP+CP: 1k + 330 pF = 482 kHz input RC; RS = source termination 1k







Tap buffers: TLV9064 followers, outputs T1_k feed the crossbar (global labels)



dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /line1/
File: line1.kicad_sch

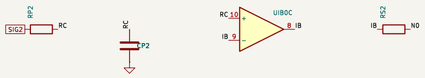
Title: Delay line 1

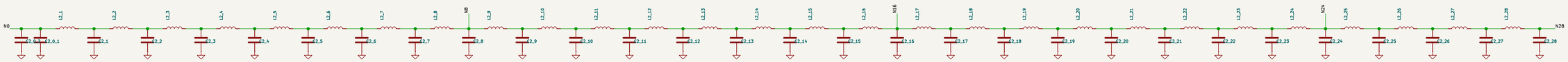
Size: A2	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 4/20

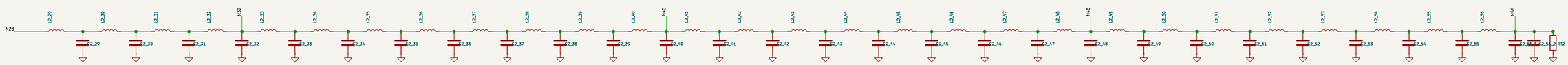
Delay line 2: 56 x (L 120 uH FHW1812IF121JST + C 120 pF C0G), Z0 = 1 kohm, T = 120 ns/section, taps k = 0..7 at sections 0, 8, ..., 56 (T_s = 8 sections = 0.96 us)

All L2_s: 120 uH 5 %, 1812; C2_n: 120 pF C0G 0603; end caps 2 x 30 pF (60 pF) at nodes 0 and 56; Rs = Rt = 1k

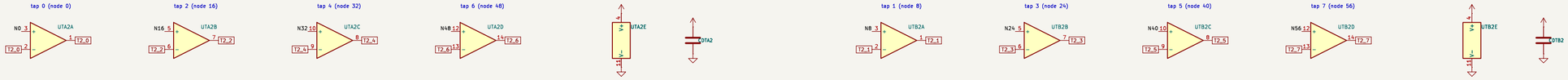
RP+CP: 1k + 330 pF = 482 kHz input RC; RS = source termination 1k







Tap buffers: TLV9064 followers, outputs T2_k feed the crossbar (global labels)



dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /line2/
File: line2.kicad_sch

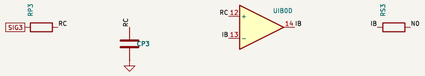
Title: Delay line 2

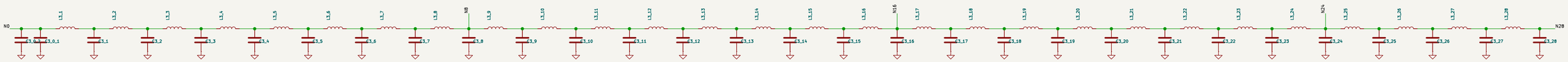
Size: A2	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 5/20

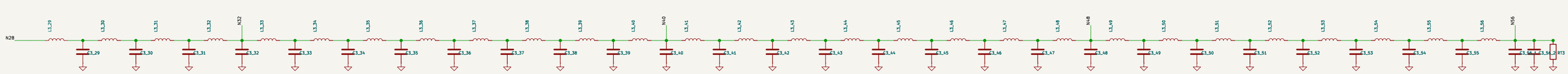
Delay line 3: 56 x (L 120 uH FHW1812IF121JST + C 120 pF C0G), Z0 = 1 kohm, T = 120 ns/section, taps k = 0..7 at sections 0, 8, ..., 56 (T_s = 8 sections = 0.96 us)

All L3_s: 120 uH 5 %, 1812; C3_n: 120 pF C0G 0603; end caps 2 x 30 pF (60 pF) at nodes 0 and 56; Rs = Rt = 1k

RP+CP: 1k + 330 pF = 482 kHz input RC; RS = source termination 1k

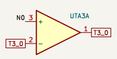




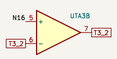


Tap buffers: TLV9064 followers, outputs T3_k feed the crossbar (global labels)

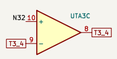
tap 0 (node 0)



tap 2 (node 16)

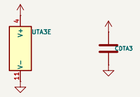


tap 4 (node 32)

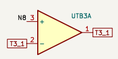


tap 6 (node 48)

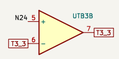




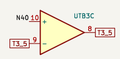
tap 1 (node 8)



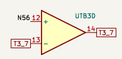
tap 3 (node 24)

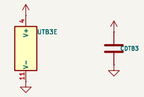


tap 5 (node 40)



tap 7 (node 56)





dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /line3/
File: line3.kicad_sch

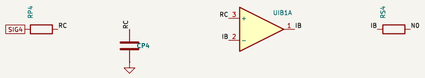
Title: Delay line 3

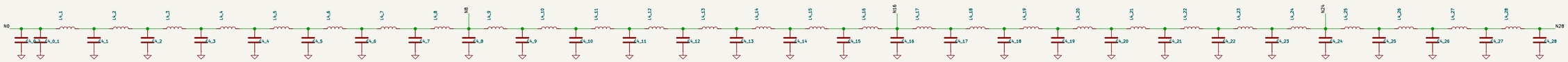
Size: A2	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 6/20

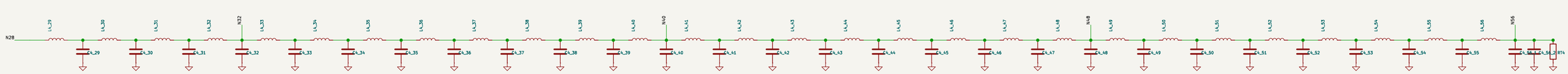
Delay line 4: 56 x (L 120 uH FHW1812IF121JST + C 120 pF C0G), Z0 = 1 kohm, T = 120 ns/section, taps k = 0..7 at sections 0, 8, ..., 56 (T_s = 8 sections = 0.96 us)

All L4_s: 120 uH 5 %, 1812; C4_n: 120 pF C0G 0603; end caps 2 x 30 pF (60 pF) at nodes 0 and 56; Rs = Rt = 1k

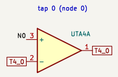
RP+CP: 1k + 330 pF = 482 kHz input RC; RS = source termination 1k

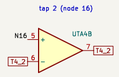


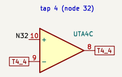


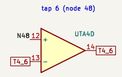


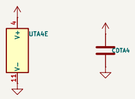
Tap buffers: TLV9064 followers, outputs T4_k feed the crossbar (global labels)

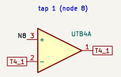


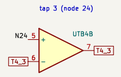


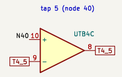


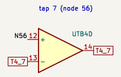


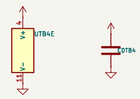












dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /line4/
File: line4.kicad_sch

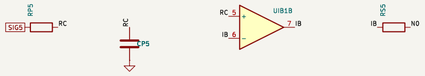
Title: Delay line 4

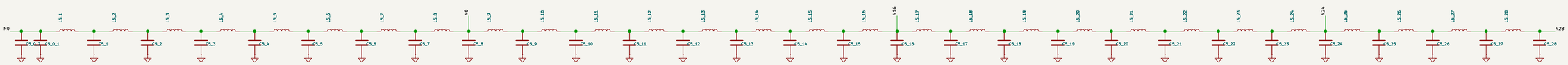
Size: A2	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 7/20

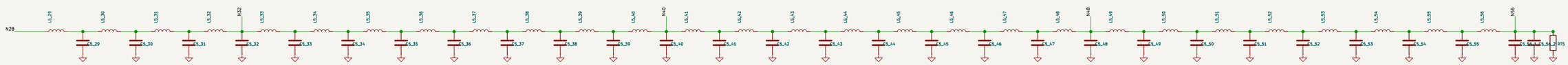
Delay line 5: 56 x (L 120 uH FHW1812IF121JST + C 120 pF C0G), Z0 = 1 kohm, T = 120 ns/section, taps k = 0..7 at sections 0, 8, ..., 56 (T_s = 8 sections = 0.96 us)

All L5_s: 120 uH 5 %, 1812; C5_n: 120 pF C0G 0603; end caps 2 x 30 pF (60 pF) at nodes 0 and 56; Rs = Rt = 1k

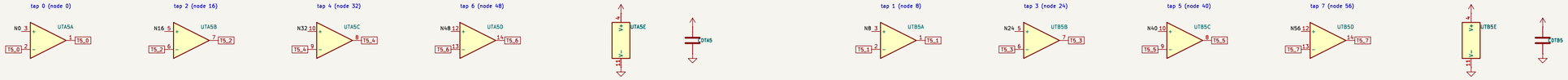
RP+CP: 1k + 330 pF = 482 kHz input RC; RS = source termination 1k







Tap buffers: TLV9064 followers, outputs T5_k feed the crossbar (global labels)



dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /line5/
File: line5.kicad_sch

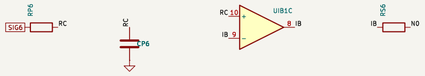
Title: Delay line 5

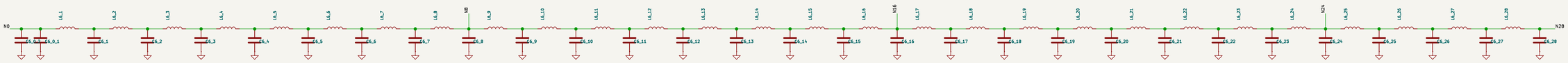
Size: A2	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 8/20

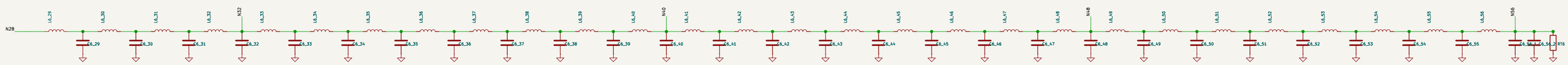
Delay line 6: 56 x (L 120 uH FHW1812IF121JST + C 120 pF C0G), Z0 = 1 kohm, T = 120 ns/section, taps k = 0..7 at sections 0, 8, ..., 56 (T_s = 8 sections = 0.96 us)

All L6_s: 120 uH 5 %, 1812; C6_n: 120 pF C0G 0603; end caps 2 x 30 pF (60 pF) at nodes 0 and 56; Rs = Rt = 1k

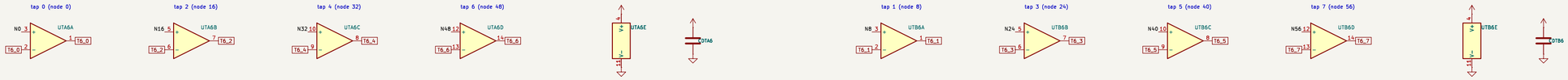
RP+CP: 1k + 330 pF = 482 kHz input RC; RS = source termination 1k







Tap buffers: TLV9064 followers, outputs T6_k feed the crossbar (global labels)



dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /line6/
File: line6.kicad_sch

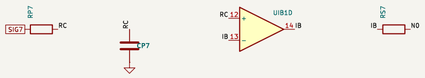
Title: Delay line 6

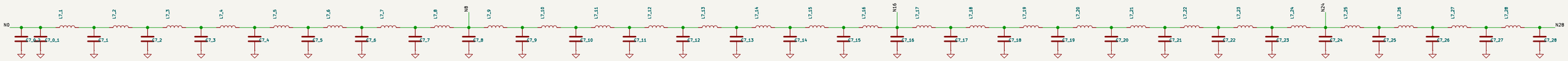
Size: A2	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 9/20

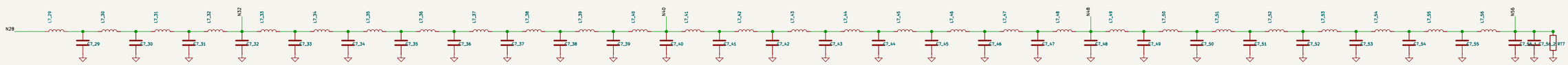
Delay line 7: 56 x (L 120 uH FHW1812IF121JST + C 120 pF C0G), Z0 = 1 kohm, T = 120 ns/section, taps k = 0..7 at sections 0, 8, ..., 56 (T_s = 8 sections = 0.96 us)

All L7_s: 120 uH 5 %, 1812; C7_n: 120 pF C0G 0603; end caps 2 x 30 pF (60 pF) at nodes 0 and 56; Rs = Rt = 1k

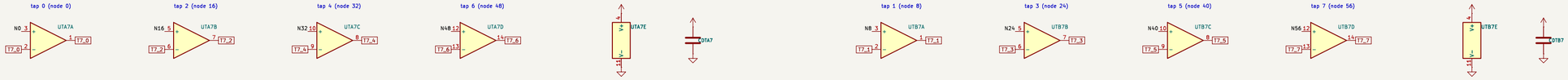
RP+CP: 1k + 330 pF = 482 kHz input RC; RS = source termination 1k







Tap buffers: TLV9064 followers, outputs T7_k feed the crossbar (global labels)



dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /line7/
File: line7.kicad_sch

Title: Delay line 7

Size: A2	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 10/20

Crossbar, layer 3: weight W[o,i,k] in {−1,0,+1}: W=+1 populate RXP_o_i_k (tap -> SN_o), W=−1 populate RXN_o_i_k (tap -> SP_o), W=0 both DNP. All 10k 1 % 0402 (C25744). The pick-and-place file is the trained model.

output 0: + -> SN0 − -> SP0

output 1: + -> SN1 − -> SP1

output 2: + -> SN2 − -> SP2

output 3: + -> SN3 − -> SP3

output 4: + -> SN4 − -> SP4

output 5: + -> SN5 − -> SP5

output 6: + -> SN6 − -> SP6

output 7: + -> SN7 − -> SP7

line 0

line 1

line 2

line 3

line 4

line 5

line 6

line 7

dinn phase 2: designed, ERC-checked, not built, not measured

Sheet: /crossbar/

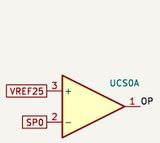
File: crossbar.kicad_sch

Title: Crossbar 0402 10k footprints

Size: A1	Date: 2026-11-03	Rev: M3a
KiCad E.D.A. 9.0.9		Id: 11/20

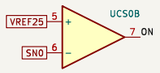
Channel 0: differential summer (SP/SN) + bias DAC, BJT diff pair (tanh), difference amp x0.525, 74HCT4066 track-and-hold + buffer -> OUT0

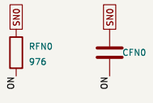
layer 3: R_f = 976 ohm, R_b = 7870 ohm, diff-amp R_f = 10500 ohm; bias -0.067 model units -> DAC 0.994 V (code 1989, 2.048 V ref); RO = 13300 ohm



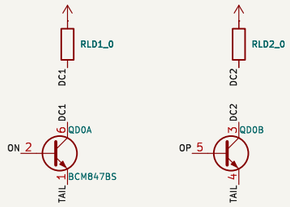


SP/SN: crossbar summing nodes (virtual ground at 2.5 V)

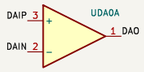










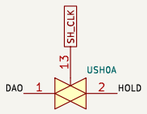


RO: bias offset current from +5VA so the 3.3 V-powered DAC (0..2.048 V) can trim around the 2.5 V virtual ground

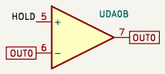


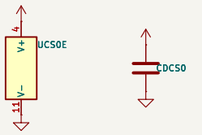
diff pair: Q1 base = ON, Q2 base = OP; tail = mirror output (MIRB from power sheet), -1 mA

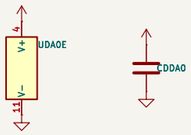
track-and-hold: 74HCT4066 + 560 pF C0G, buffer -> OUT (ribbon SIG, 2.5 V +- 0.5 V PAM into the next tile)

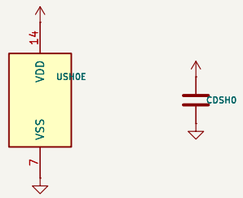










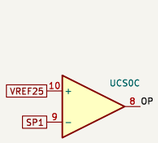


dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /ch0/		
File: ch0.kicad_sch		
Title: Channel 0		
Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 12/20

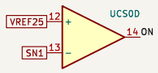
Channel 1: differential summer (SP/SN) + bias DAC, BJT diff pair (tanh), difference amp x0.525, 74HCT4066 track-and-hold + buffer -> OUT1

layer 3: R_f = 976 ohm, R_b = 7870 ohm, diff-amp R_f = 10500 ohm; bias -2.019 model units -> DAC 0.226 V (code 452, 2.048 V ref); RO = 13300 ohm





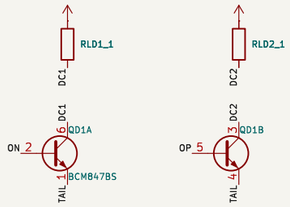
SP/SN: crossbar summing nodes (virtual ground at 2.5 V)



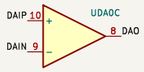




RO: bias offset current from +5VA so the 3.3 V-powered DAC (0..2.048 V) can trim around the 2.5 V virtual ground



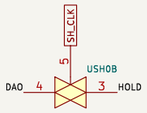




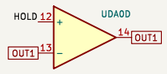


diff pair: Q1 base = ON, Q2 base = OP; tail = mirror output (MIRB from power sheet), ~1 mA

track-and-hold: 74HCT4066 + 560 pF C0G, buffer -> OUT (ribbon SIG, 2.5 V +- 0.5 V PAM into the next tile)







dlnn phase 2: designed, ERC-checked, not built, not measured

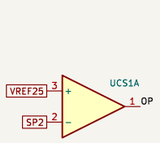
Sheet: /ch1/
File: ch1.kicad_sch

Title: Channel 1

Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 13/20

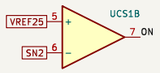
Channel 2: differential summer (SP/SN) + bias DAC, BJT diff pair (tanh), difference amp x0.525, 74HCT4066 track-and-hold + buffer -> OUT2

layer 3: R_f = 976 ohm, R_b = 7870 ohm, diff-amp R_f = 10500 ohm; bias -0.283 model units -> DAC 0.909 V (code 1819, 2.048 V ref); RO = 13300 ohm

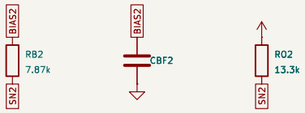


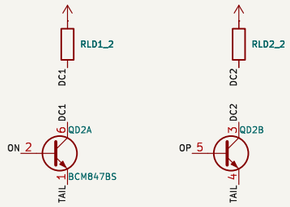


SP/SN: crossbar summing nodes (virtual ground at 2.5 V)

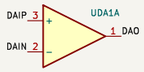










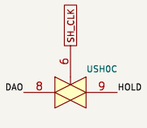


RO: bias offset current from +5VA so the 3.3 V-powered DAC (0..2.048 V) can trim around the 2.5 V virtual ground

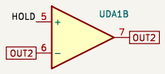


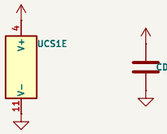
diff pair: Q1 base = ON, Q2 base = OP; tail = mirror output (MIRB from power sheet), -1 mA

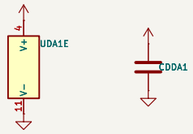
track-and-hold: 74HCT4066 + 560 pF C0G, buffer -> OUT (ribbon SIG, 2.5 V +- 0.5 V PAM into the next tile)









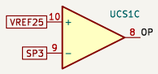


dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /ch2/		
File: ch2.kicad_sch		
Title: Channel 2		
Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 14/20

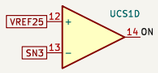
Channel 3: differential summer (SP/SN) + bias DAC, BJT diff pair (tanh), difference amp x0.525, 74HCT4066 track-and-hold + buffer -> OUT3

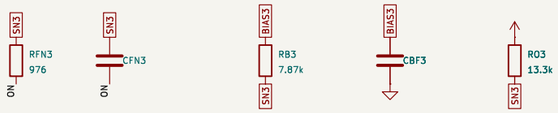
layer 3: R_f = 976 ohm, R_b = 7870 ohm, diff-amp R_f = 10500 ohm; bias +0.348 model units -> DAC 1.157 V (code 2315, 2.048 V ref); RO = 13300 ohm

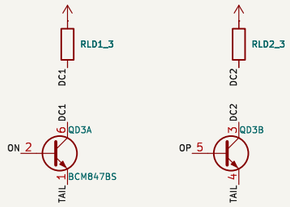




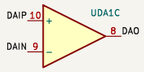
SP/SN: crossbar summing nodes (virtual ground at 2.5 V)









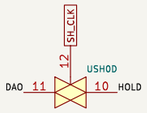


RO: bias offset current from +5VA so the 3.3 V-powered DAC (0..2.048 V) can trim around the 2.5 V virtual ground

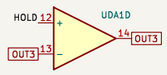


diff pair: Q1 base = ON, Q2 base = OP; tail = mirror output (MIRB from power sheet), -1 mA

track-and-hold: 74HCT4066 + 560 pF C0G, buffer -> OUT (ribbon SIG, 2.5 V +- 0.5 V PAM into the next tile)





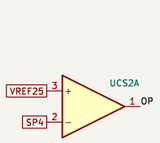


dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /ch3/		
File: ch3.kicad_sch		
Title: Channel 3		
Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 15/20

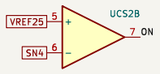
Channel 4: differential summer (SP/SN) + bias DAC, BJT diff pair (tanh), difference amp x0.525, 74HCT4066 track-and-hold + buffer -> OUT4

layer 3: R_f = 976 ohm, R_b = 7870 ohm, diff-amp R_f = 10500 ohm; bias +1.411 model units -> DAC 1.576 V (code 3152, 2.048 V ref); RO = 13300 ohm





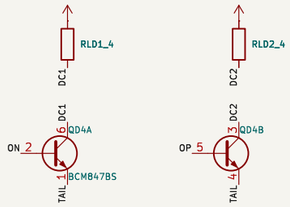
SP/SN: crossbar summing nodes (virtual ground at 2.5 V)



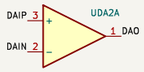




RO: bias offset current from +5VA so the 3.3 V-powered DAC (0..2.048 V) can trim around the 2.5 V virtual ground



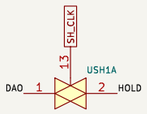




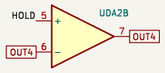


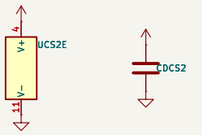
diff pair: Q1 base = ON, Q2 base = OP; tail = mirror output (MIRB from power sheet), ~1 mA

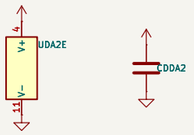
track-and-hold: 74HCT4066 + 560 pF C0G, buffer -> OUT (ribbon SIG, 2.5 V +- 0.5 V PAM into the next tile)

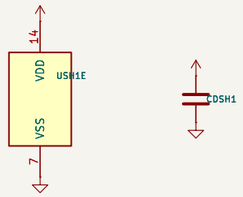












dlnn phase 2: designed, ERC-checked, not built, not measured

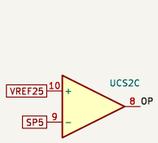
Sheet: /ch4/
File: ch4.kicad_sch

Title: Channel 4

Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 16/20

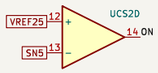
Channel 5: differential summer (SP/SN) + bias DAC, BJT diff pair (tanh), difference amp x0.525, 74HCT4066 track-and-hold + buffer -> OUT5

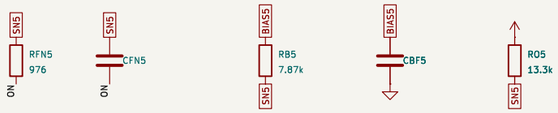
layer 3: R_f = 976 ohm, R_b = 7870 ohm, diff-amp R_f = 10500 ohm; bias +0.158 model units -> DAC 1.083 V (code 2166, 2.048 V ref); RO = 13300 ohm

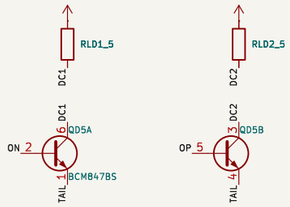




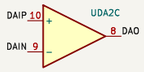
SP/SN: crossbar summing nodes (virtual ground at 2.5 V)









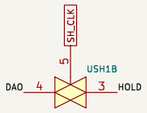


RO: bias offset current from +5VA so the 3.3 V-powered DAC (0..2.048 V) can trim around the 2.5 V virtual ground

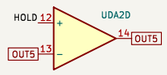


diff pair: Q1 base = ON, Q2 base = OP; tail = mirror output (MIRB from power sheet), -1 mA

track-and-hold: 74HCT4066 + 560 pF C0G, buffer -> OUT (ribbon SIG, 2.5 V +- 0.5 V PAM into the next tile)





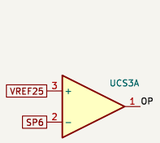


dlnn phase 2: designed, ERC-checked, not built, not measured

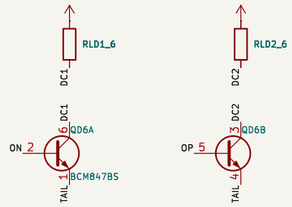
Sheet: /ch5/		
File: ch5.kicad_sch		
Title: Channel 5		
Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 17/20

Channel 6: differential summer (SP/SN) + bias DAC, BJT diff pair (tanh), difference amp x0.525, 74HCT4066 track-and-hold + buffer -> OUT6

layer 3: R_f = 976 ohm, R_b = 7870 ohm, diff-amp R_f = 10500 ohm; bias +1.024 model units -> DAC 1.424 V (code 2847, 2.048 V ref); RO = 13300 ohm

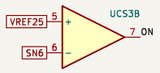






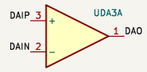


SP/SN: crossbar summing nodes (virtual ground at 2.5 V)







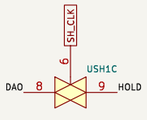


RO: bias offset current from +5VA so the 3.3 V-powered DAC (0..2.048 V) can trim around the 2.5 V virtual ground

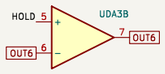


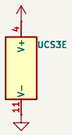
diff pair: Q1 base = ON, Q2 base = OP; tail = mirror output (MIRB from power sheet), -1 mA

track-and-hold: 74HCT4066 + 560 pF C0G, buffer -> OUT (ribbon SIG, 2.5 V +- 0.5 V PAM into the next tile)

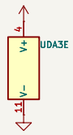












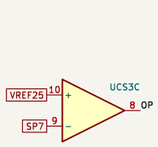


dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /ch6/		
File: ch6.kicad_sch		
Title: Channel 6		
Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 18/20

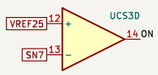
Channel 7: differential summer (SP/SN) + bias DAC, BJT diff pair (tanh), difference amp x0.525, 74HCT4066 track-and-hold + buffer -> OUT7

layer 3: R_f = 976 ohm, R_b = 7870 ohm, diff-amp R_f = 10500 ohm; bias -0.321 model units -> DAC 0.895 V (code 1789, 2.048 V ref); RO = 13300 ohm





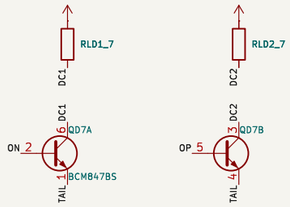
SP/SN: crossbar summing nodes (virtual ground at 2.5 V)



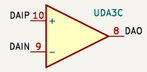




RO: bias offset current from +5VA so the 3.3 V-powered DAC (0..2.048 V) can trim around the 2.5 V virtual ground



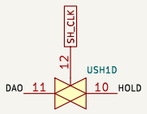




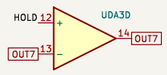


diff pair: Q1 base = ON, Q2 base = OP; tail = mirror output (MIRB from power sheet), -1 mA

track-and-hold: 74HCT4066 + 560 pF C0G, buffer -> OUT (ribbon SIG, 2.5 V +- 0.5 V PAM into the next tile)





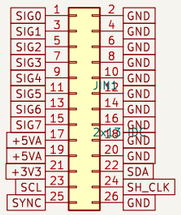


dlnn phase 2: designed, ERC-checked, not built, not measured

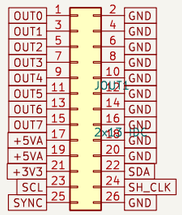
Sheet: /ch7/		
File: ch7.kicad_sch		
Title: Channel 7		
Size: A3	Date: 2026-11-03	**Rev: M3a**
KiCad E.D.A. 9.0.9		Id: 19/20

Connectors: J_IN / J_OUT 2x13 IDC ribbon (<= 10 cm), J_PS 2x8 partial-sum header (DNP at width 8), 2 x MCP4728 bias DACs, tile-ID jumpers, monitor tap

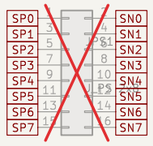
J_IN



J_OUT



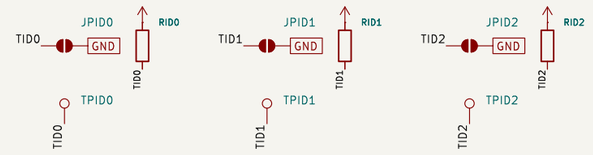
J_PS (DNP, stitching only; pins = SP0/SN0 ... SP7/SN7 summing nodes)



monitor: line 0, tap 7 (T0_7) for clock calibration

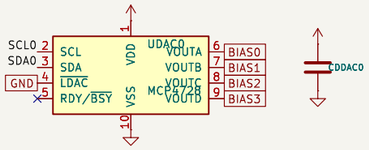


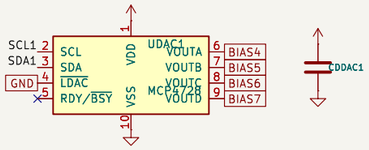
tile ID = TCA9548A address: 3 solder jumpers to GND (closed = 0), 10k pull-ups to +3V3 (open = 1); address 0x70 + ID. Test

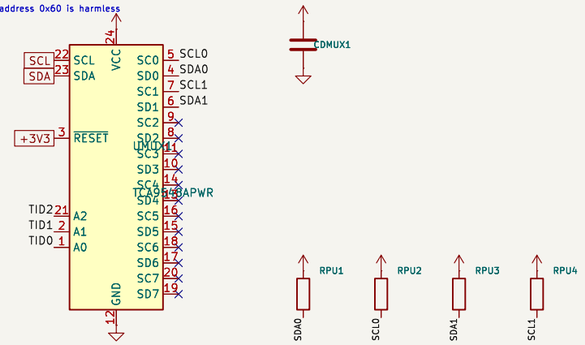


TCA9548A per tile: host SDA/SCL (ribbon, pulled up on the carrier) -> ch0 -> UDAC0, ch1 -> UDAC1; ch2..7 unused. RESET tied to +3V3. Channel pull-

UDAC0: bias outputs BIAS0..BIAS3; VDD = 3.3 V, internal 2.048 V reference, gain x1, centre code 2048 = 1.024 V; LDAC tied low; UDAC1 on bias outputs BIAS4..BIAS7; shared default address 0x60 is harmless







Bias DAC range (fix of issue 2): see channel sheets, RO from +5VA offsets the 0..2.048 V DAC range to the summer virtual ground

dlnn phase 2: designed, ERC-checked, not built, not measured

Sheet: /connectors/
File: connectors.kicad_sch

Title: Connectors, bias DACs, ID, monitor

Size: A3	Date: 2026-11-03	Rev: M3a
KiCad E.D.A. 9.0.9		Id: 20/20